KEY TOPIC 4 • REVISION & PRACTICE COMPANION

Medicine in Modern Britain

c1900–present: Ideas on Causes, Diagnostic Technology, The NHS, Penicillin & Lung Cancer



Primary Visual Evidence: *Health Minister Aneurin Bevan visiting Sylvia Beckingham (first NHS patient), Park Hospital, Davyhulme, 5 July 1948* CENTRAL OFFICE OF INFORMATION ARCHIVE

PUPIL NAME: _____ **TARGET:** _____ **CLASS:** _____

KEY TOPIC 4 SPECIFICATION FRAMEWORK (1HI0/11)	LESSON DONE	QUIZ [10]
Key Topic 4.1: Ideas on Causes: Genetics, DNA & The Human Genome Project *How did Rosalind Franklin's Photograph 51, Watson & Crick, and genomic sequencing unlock hereditary disease?*	☐	☐
Key Topic 4.2: Lifestyle Factors & The Technological Revolution in Diagnosis *How did the shift to chronic lifestyle illnesses, Röntgen's X-rays, and Hounsfield's CT scanner eliminate exploratory surgery?*	☐	☐
Key Topic 4.3: Magic Bullets, High-Tech Treatments & The Birth of the NHS *How did Salvarsan 606, Prontosil, and Aneurin Bevan's 1948 NHS create free healthcare for every British citizen?*	☐	☐
Key Topic 4.4: Case Study 1: The Antibiotic Revolution: Fleming, Florey & Chain *How did Fleming's 1928 discovery, the Oxford team's purification, and American deep-tank vats deliver 2.3m doses for D-Day?*	☐	☐
Key Topic 4.5: Case Study 2: Public Health & The Fight Against Lung Cancer *How did Doll & Hill prove the tobacco link, and how did ad bans, the 2007 public smoking ban, and plain packaging end laissez-faire?*	☐	☐

Personal revision tracking ledger: ensure every specification bullet point is mastered.

LIVING TIMELINE • PART 1: MAGIC BULLETS, BLOOD GROUPS & EARLY ANTIBIOTICS (1900–1940)

Instructions: Complete the timeline sketches and notes as you master each enquiry lesson.

1901 • Karl Landsteiner: ABO Blood Groups Discovered

Key Topic 4.1

Landsteiner identifies A, B, and O blood groups, proving agglutination occurs when incompatible blood types mix. This discovery transforms blood transfusions from fatal gambles into safe, predictable surgical lifelines, foundational for modern trauma surgery.

1909 • Paul Ehrlich & Sahachiro Hata: Salvarsan 606

Key Topic 4.1

Building on dye research, Ehrlich and Hata systematically test hundreds of arsenic compounds, discovering arsphenamine (compound 606). Humanity's first synthetic chemical 'magic bullet' targets syphilis bacteria without poisoning human tissue.

1928 • Alexander Fleming: Accidental Discovery of Penicillin

Key Topic 4.2

Returning from holiday, Fleming notices a contaminated staphylococcus culture dish with a halo of destroyed bacteria around Penicillium notatum mould. Fleming publishes in 1929, though unable to purify or extract enough active substance for clinical trials.

1932 • Gerhard Domagk: Prontosil & Sulfonamide Drugs

Key Topic 4.1

Domagk demonstrates that the red leather dye Prontosil cures streptococcal blood poisoning in mice (and saves his own daughter's arm). Chemists isolate the active ingredient sulfonamide, launching the world's first mass-prescribed class of antibacterial wonder drugs.

The 20th century transformed pharmacology from toxic general poisons into targeted magic bullets.

LIVING TIMELINE • PART 2: MASS ANTIBIOTICS, THE NHS, DNA & ONCOLOGY (1940–PRESENT)

Instructions: Complete the timeline sketches and notes as you master each enquiry lesson.

1941–44 • Florey, Chain & Heatley: Penicillin Mass Production

Key Topic 4.2

Howard Florey and Ernst Chain purify penicillin at Oxford, successfully treating mice and Albert Alexander (1941). Heatley and Florey travel to Peoria, Illinois, using corn-steep liquor and deep-tank fermentation to produce 2.3 million doses for D-Day casualties.

1948 • Aneurin Bevan: Foundation of the NHS

Key Topic 4.3

On 5 July 1948, the National Health Service opens at Park Hospital, Manchester. Championed by Bevan and the 1942 Beveridge Report, it nationalises voluntary and municipal hospitals, providing universal healthcare free at the point of clinical delivery.

1953 • Watson, Crick & Franklin: DNA Double Helix Structure

Key Topic 4.4

Using Rosalind Franklin and Maurice Wilkins's X-ray crystallography (Photograph 51), Watson and Crick deduce the double-helix structure of DNA at Cambridge, unlocking hereditary genetic code and modern molecular gene therapy.

1971–2003 • CT Scanners (1971) & The Human Genome Project (2003)

Key Topic 4.4 • 4.5

Godfrey Hounsfield invents the EMI CT scanner, combining computer algorithms with X-rays to image internal organs without surgery. In 2003, the international Human Genome Project maps all 3 billion chemical base pairs in human DNA, founding precision medicine.

Why did the discovery of DNA revolutionize understanding of disease causes, yet take decades to yield treatments?

CHRONOLOGICAL INQUIRY SPINE • CORE CAUSAL NARRATIVE

Take precise, structured notes alongside each milestone as your teacher narrates the history

1 **c1900–1940s**

Mendel & Early Genetics

- Growing wrinkled green garden peas
- Spotting dark rod shapes inside cells
- Assuming heavy protein chains hold secrets

FOCUS CLUE
Why did doctors assume proteins carried hereditary traits?

2 **1951–1952**

Franklin & Photograph 51

- Beaming invisible rays through wet DNA fibers
- Exposing glass photographic plates for sixty hours
- Revealing clear fuzzy X-shaped shadow marks

FOCUS CLUE
How did Photograph 51 prove DNA's helical structure?

3 **1953**

Watson & Crick's Model

- Twisting thin metal wire into spirals
- Locking four chemical pairs like puzzle pieces
- Rushing to celebratory pints at The Eagle pub

FOCUS CLUE
How did base pairing explain hereditary replication?

4 **1990–2003**

Human Genome Project

- Linking laboratory supercomputers across six nations
- Reading three billion chemical letters in blood
- Publishing complete human code free online

FOCUS CLUE
Why was mapping 3 billion base pairs a medical milestone?

5 **2000s–Present**

Screening & Treatment Gap

- Testing saliva drops for faulty breast genes
- Matching targeted pills to personal body code
- Struggling to fix broken genetic code inside bodies

FOCUS CLUE
Why did understanding causes outpace finding cures?

Franklin's 62-hour X-ray exposure revealed the double helix; Watson and Crick celebrated in The Eagle pub.

Question 4: Explain Why [12 marks • 15 mins]

EXTENDED WRITING ASSESSMENT

Explain why there was rapid progress in understanding the causes of disease in the period c1950 to the present. [12 marks]

Stimulus: Rosalind Franklin's Photograph 51 (1952) | The Human Genome Project (1990–2003) *(You must also use information of your own)*

ACCESSIBLE WRITING FRAMES:
One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...

KEY VOCABULARY BANK:
Gregor Mendel • Rosalind Franklin • Photograph 51 (1952) • Watson & Crick • double helix model • Human Genome Project (1990–2003) • gene screening • hereditary disease

Write your response in the space provided below:

(Response continues on facing page)

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 4.1). In the drawing box, sketch Franklin's Photograph 51 cross pattern and Watson & Crick's double helix model. Annotate how base pairing unlocked the causes of hereditary disease! **[ ] Done**

Continue your response to Question 4 below:

Final analytical evaluation & concluding historical verdict:

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** [4 MARKS • 5 MINS]

Explain one way in which scientific methods for investigating disease causes in the late twentieth century (c1950–c2000) were different from methods used in the late nineteenth century (c1870–c1900). [4 marks]

Were high-tech diagnostic scanners more vital to modern health than state lifestyle campaigns?

CHRONOLOGICAL INQUIRY SPINE • CORE CAUSAL NARRATIVE

Take precise, structured notes alongside each milestone as your teacher narrates the history

1 **1895–1918**

Röntgen & Early X-Rays

- Photographing wife Bertha's hand and gold wedding ring
- Driving mobile medical vans to muddy trenches
- Finding jagged lead shrapnel without deep cuts

FOCUS CLUE
How did X-rays transform surgical trauma diagnosis in WWI?

2 **1930s–1960s**

Blood Labs & Endoscopy

- Spinning red test tubes in fast centrifuges
- Counting abnormal white cells on glass slides
- Sliding flexible glass cables down throat tubes

FOCUS CLUE
Why was fiber-optic endoscopy superior to exploratory surgery?

3 **1971–1980s**

Hounsfield's CT & MRI

- Spinning heavy donut scanners around hospital beds
- Combining computer pictures into 3D body slices
- Using magnetic radio waves to find brain lumps

FOCUS CLUE
Why did CT scans transform internal soft-tissue diagnosis?

4 **c1950–1980s**

Epidemiological Shift

- Banishing deadly childhood fevers with wonder drugs
- People living past seventy years of age
- Rising chest tumours and clogged heart arteries

FOCUS CLUE
What drove the shift from infectious to lifestyle diseases?

5 **1980s–Present**

State Lifestyle Campaigns

- Clicking compulsory car seatbelts across chests
- Extinguishing glowing cigarettes outside pub doors
- Taxing sugary fizzy cans and advertising fruit

FOCUS CLUE
Why do governments favour lifestyle prevention over treatment?

Bertha Röntgen cried "I have seen my death!" upon seeing her bones; doctors saw the dawn of non-invasive diagnosis.

Question 5/6: Evaluative Essay [16+4 marks • 20 mins]

EXTENDED WRITING ASSESSMENT

'Technological innovations in medical diagnosis were more important than public health lifestyle campaigns in improving health in Britain c1900–present.' How far do you agree? [16+4 marks]

Stimulus: Wilhelm Röntgen's X-rays (1895) | The 2007 smoking ban in public places *(You must also use information of your own)*

ACCESSIBLE WRITING FRAMES:

On the one hand, high-tech diagnosis was vital because... • For example, ... • This was significant because... • On the other hand, lifestyle campaigns mattered because... • In particular, ... • Overall, the most decisive factor was... because...

KEY VOCABULARY BANK:

Wilhelm Röntgen (1895) • X-rays • CT scanner (1971) • Godfrey Hounsfield • MRI • non-invasive imaging • 2007 smoking ban • Change4Life

Write your response in the space provided below:

(Response continues on facing page)

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 4.2). In the drawing box, sketch Bertha Röntgen's hand X-ray and Godfrey Hounsfield's circular CT scanner. Annotate how scanning permanently ended blind exploratory surgery! **[ ] Done**

Continue your response to Question 5/6 below:

Final analytical evaluation & concluding historical verdict:

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** [4 MARKS • 5 MINS]

Explain one way in which methods of diagnosing illness in the late twentieth century (c1970–present) were different from methods used in the nineteenth century (c1800–c1900). [4 marks]

Was the establishment of the NHS in 1948 more transformative than the development of magic bullets?

CHRONOLOGICAL INQUIRY SPINE • CORE CAUSAL NARRATIVE

Take precise, structured notes alongside each milestone as your teacher narrates the history

1 **1906–1909**

Ehrlich & Salvarsan 606

- Testing hundreds of toxic yellow arsenic mixtures
- Hunting chemical bullets that only hit germs
- Curing deadly human syphilis with magic bullet

FOCUS CLUE
Why was Salvarsan 606 termed the first "magic bullet"?

2 **1932–1935**

Domagk & Prontosil

- Injecting bright red leather dye into mice
- Saving sick daughter from arm-amputation death
- Discovering active white sulfa medicine powder

FOCUS CLUE
How did Prontosil prove synthetic drugs could kill microbes?

3 **1942**

The Beveridge Report

- Listing Five Giants that crush poor workers
- Surviving Nazi air bombs with community spirit
- Demanding free medical care for every citizen

FOCUS CLUE
How did WWII unity create momentum for state healthcare?

4 **1945–1948**

Bevan & Launch of the NHS

- Offering private bed perks to stubborn doctors
- Delivering free spectacles and dentures to families
- Abolishing terrifying doctor bills forever

FOCUS CLUE
How did Bevan compromise to persuade resistant doctors?

5 **1950s–Present**

High-Tech Care & Costs

- Pumping blood through giant artificial kidney filters
- Replacing worn hip joints with metal balls
- Charging small fees for paper medicine slips

FOCUS CLUE
Why has modern medical success created financial strain?

Ehrlich tested 605 failed arsenic compounds before compound 606 finally cured syphilis in 1909.

Question 5/6: Evaluative Essay [16+4 marks • 20 mins]

EXTENDED WRITING ASSESSMENT

'The establishment of the National Health Service in 1948 was the most significant turning point in medical care in the period c1900–present.' How far do you agree? [16+4 marks]

Stimulus: The National Health Service (1948) | The discovery of Salvarsan 606 (1909) *(You must also use information of your own)*

ACCESSIBLE WRITING FRAMES:

On the one hand, the NHS was transformative because... • For example, ... • This was significant because... • On the other hand, magic bullets were crucial because... • In particular, ... • Overall, the most decisive factor was... because...

KEY VOCABULARY BANK:

Paul Ehrlich • Salvarsan 606 (1909) • Gerhard Domagk • Prontosil (1932) • William Beveridge (1942) • Aneurin Bevan • NHS (1948) • free at point of delivery

Write your response in the space provided below:

(Response continues on facing page)

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 4.3). In the drawing box, sketch Ehrlich's test tube of Salvarsan 606 and the 1948 NHS information leaflet. Annotate how free healthcare abolished medical poverty! [ ] Done

Continue your response to Question 5/6 below:

Final analytical evaluation & concluding historical verdict:

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** [4 MARKS • 5 MINS]

Explain one way in which healthcare access in Britain after 1948 was different from healthcare access in the 1930s. [4 marks]

Why did penicillin require a global war and American industrial might to become a mass-produced cure?

CHRONOLOGICAL INQUIRY SPINE • CORE CAUSAL NARRATIVE

Take precise, structured notes alongside each milestone as your teacher narrates the history

1 **1928–1929**

Fleming's Discovery

- Finding fluffy mould on dirty glass dish
- Golden germs dissolving in clear circle halos
- Failing to separate fragile liquid juice

FOCUS CLUE
Why could Fleming not turn mould into a clinical medicine?

2 **1938–1940**

Florey & Chain at Oxford

- Brewing fungus inside hospital bedpans and milk churns
- Winning American dollar grants during air raids
- Freezing brown liquid into precious yellow crystals

FOCUS CLUE
How did Norman Heatley solve the purification challenge?

3 **1940–1941**

Mice Trials & Albert Alexander

- Infecting eight mice and keeping treated four alive
- Scraping rose thorn scratches on policeman's face
- Running out of drops before patient recovers

FOCUS CLUE
Why was Alexander's trial both a success and a tragedy?

4 **1941–1944**

Peoria & US Mass Production

- Feeding corn beer wash to rotting melon mould
- Pumping sterile air through giant ten-thousand-gallon vats
- Brewing million doses for D-Day landing craft

FOCUS CLUE
How did US industrial fermentation enable mass production?

5 **1944–Present**

D-Day & Superbugs

- Spraying white powder directly into torn soldiers' limbs
- Three scientists sharing golden Nobel prize medal
- Overusing capsules creating tough hospital superbugs

FOCUS CLUE
Why did Fleming warn that antibiotic overuse causes resistance?

Fleming almost washed his petri dish in Lysol, but noticed a clear halo where mould destroyed golden staphylococci.

Question 4: Explain Why [12 marks • 15 mins]

EXTENDED WRITING ASSESSMENT

Explain why penicillin was successfully developed and mass-produced in the period 1928–1945. [12 marks]

Stimulus: Alexander Fleming's discovery (1928) | The impact of the Second World War *(You must also use information of your own)*

ACCESSIBLE WRITING FRAMES:

One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...

KEY VOCABULARY BANK:

Alexander Fleming (1928) • penicillin • Howard Florey • Ernst Chain • Norman Heatley • Albert Alexander (1941) • Peoria fermentation • mass production (1944)

Write your response in the space provided below:

(Response continues on facing page)

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 4.4). In the drawing box, sketch Fleming's contaminated petri dish and an American 10,000-gallon deep-tank vat. Annotate how penicillin saved Allied troops on D-Day! **[ ] Done**

Continue your response to Question 4 below:

Final analytical evaluation & concluding historical verdict:

• **QUESTION 3: EXPLAIN ONE SIMILARITY [4 MARKS]** [4 MARKS • 5 MINS]

Explain one way in which Alexander Fleming's discovery of penicillin in 1928 was similar to Edward Jenner's discovery of the smallpox vaccine in 1796. [4 marks]

Why has the modern fight against lung cancer required state coercion rather than laboratory cures?

CHRONOLOGICAL INQUIRY SPINE • CORE CAUSAL NARRATIVE

Take precise, structured notes alongside each milestone as your teacher narrates the history

1 **1920s–1950**

The Mysterious Epidemic

- Men coughing black blood into hospital basins
- Blaming smelly road tarmac and lorry exhaust fumes
- Colourful billboards showing smiling smoking doctors

FOCUS CLUE
Why did early doctors fail to link smoking to lung cancer?

2 **1950–1954**

Doll & Hill's Proof

- Interviewing sick patients across twenty London wards
- Tracking forty thousand real family doctors over years
- Proving heavy smokers face fifty-times higher risk

FOCUS CLUE
How did statistical epidemiology prove the causal link?

3 **1962–1971**

Royal College & Ad Bans

- Banning glowing cigarette commercials on evening television
- Printing bold black warning text on cardboard cartons
- Sponsoring fast racing cars to keep brands famous

FOCUS CLUE
Why did government action encounter strong tobacco lobbying?

4 **2000s**

Public Smoking Ban

- Breathing toxic secondhand smoke in crowded rooms
- Padlocking pub ashtrays and stepping outside into cold
- Raising legal counter buying age to eighteen

FOCUS CLUE
Why was passive smoking crucial in banning public smoking?

5 **2010s–Present**

Plain Packs & Oncology

- Selling plain ugly mud-green packets with graphic photos
- Guiding tiny robotic camera pincers between chest ribs
- Boosting immune soldier cells to attack hidden tumours

FOCUS CLUE
Why does low cancer survival prove prevention beats cure?

Doll and Hill proved smoking caused lung cancer in 1950; Doll immediately extinguished his pipe and lived to age 92.

Question 5/6: Evaluative Essay [16+4 marks • 20 mins]

EXTENDED WRITING ASSESSMENT

'Government public health legislation was the main reason for progress in combating lung cancer in the period c1950 to the present.' How far do you agree? [16+4 marks]

Stimulus: Doll and Hill's research (1950) | Technological treatments (radiotherapy, surgery, immunotherapy) *(You must also use information of your own)*

ACCESSIBLE WRITING FRAMES:

On the one hand, government legislation was decisive because... • For example, ... • This was significant because... • On the other hand, high-tech treatments mattered because... • In particular, ... • Overall, the most decisive factor was... because...

KEY VOCABULARY BANK:

Richard Doll • Austin Bradford Hill (1950) • British Doctors Study • 1965 TV ad ban • 2007 public smoking ban • plain packaging (2016) • radiotherapy • immunotherapy

Write your response in the space provided below:

(Response continues on facing page)

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 4.5). In the drawing box, sketch a 1950s doctor cigarette advert and a 2016 olive-green plain pack. Annotate how state legislation slashed smoking from 65% to under 13%! **[ ] Done**

Continue your response to Question 5/6 below:

Final analytical evaluation & concluding historical verdict:

• **QUESTION 3: EXPLAIN ONE SIMILARITY [4 MARKS]** [4 MARKS • 5 MINS]

Explain one way in which government intervention against lung cancer after 1965 was similar to government action in the 1875 Public Health Act. [4 marks]

PUPIL NAME: _____ TARGET: _____ CURRENT GRADE: _____

ENQUIRY	ASSESSMENT COMPONENT FOCUS	PAGE	SCORE	TEACHER COMMENT & NEXT STEPS
KT4.1	**Q4:** Explain Why Rapid Progress in Causes c1950–Present	p. 5–7	/ 12	MARKING FEEDBACK • TARGETS:
	Q3: Explain One Difference (Investigating Causes: 1880s vs 1990s)	p. 7	/ 4	
KT4.2	**Q5/6:** Evaluative Essay on Diagnostic Scanners vs Lifestyle Campaigns	p. 9–11	/ 20	MARKING FEEDBACK • TARGETS:
	Q3: Explain One Difference (Diagnostic Methods: 19th c vs Late 20th c)	p. 11	/ 4	
KT4.3	**Q5/6:** Evaluative Essay on the NHS as a Medical Turning Point (1948)	p. 13–15	/ 20	MARKING FEEDBACK • TARGETS:
	Q3: Explain One Difference (Healthcare Access: 1930s vs Post-1948)	p. 15	/ 4	
KT4.4	**Q4:** Explain Why Penicillin was Mass-Produced (1928–1945)	p. 17–19	/ 12	MARKING FEEDBACK • TARGETS:
	Q3: Explain One Similarity (Fleming 1928 vs Jenner 1796)	p. 19	/ 4	
KT4.5	**Q5/6:** Evaluative Essay on Government Action & Lung Cancer	p. 21–23	/ 20	MARKING FEEDBACK • TARGETS:
	Q3: Explain One Similarity (Anti-Smoking Laws vs 1875 Public Health Act)	p. 23	/ 4	
TOTAL	**Key Topic 4 Cumulative Examination Assessment Portfolio (10 Tasks)**	p. 4–23	/ 104	**Overall Grade:** _____ Teacher Target: _____

• DIGITAL LEARNING & RETRIEVAL HUB • INTERACTIVE LESSONS & QUIZZES — Interactive Textbook • 10 Questions Per Enquiry

KT4.1	KT4.2	KT4.3	KT4.4	KT4.5
DNA & Genetics	Lifestyle & Scans	Magic Bullets & NHS	Penicillin Study	Lung Cancer Study
LESSON HUB	LESSON HUB	LESSON HUB	LESSON HUB	LESSON HUB
MASTERY QUIZ	MASTERY QUIZ	MASTERY QUIZ	MASTERY QUIZ	MASTERY QUIZ
Score: [/ 10]	Score: [/ 10]	Score: [/ 10]	Score: [/ 10]	Score: [/ 10]

From DNA to the NHS, state action and laboratory science revolutionized healthcare; revision guarantees your GCSE success.